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Technical Assessment for .NET

Full-Stack Employee & Family Registry

Date: 6 Mar, 2026

Last Submission Date: 7 Mar, 2026

1. Project Overview

This is a basic Employee Management System designed specifically for the Bangladesh context. You are tasked with building a full-stack system that manages employee profiles and their immediate family details, while providing specific reporting capabilities.

2. Core Requirements

Feature	Description	Requirement
Database	PostgreSQL	Use EF Core with Migrations. Include SeedData for 10 initial employees.
Employee Data	Profile Management	Fields: Name, NID (10 or 17 digits), Phone (BD Format), Department, Basic Salary.
Family Data	Relationships	Support adding Spouse (Name, NID) and Children (Name, DoB) for each employee.
Search	Global Search	A single input field filtering by Name, NID, or Department (Case-insensitive).
Export (PDF)	Reporting	1. Table View: PDF of the current filtered list. 2. Individual CV: PDF summary of one employee + their family details.
Roles	Basic Auth	Viewer: Read-only access. Admin: Can Create, Update, and Delete records.
SRS Document	Documentation	A 3-5 page PDF/Markdown outlining the system architecture and edge cases.

3. Technical Tasks & Instructions

Phase 1: Database & Backend (.NET 10+)

- **Models:** Create an Employee entity with a **One-to-One** relationship for a Spouse and a **One-to-Many** relationship for Children.
- **Seeding:** Ensure the application automatically seeds the database on the first run with realistic Bangladeshi names (e.g., *Hasan, Moushumi, Tanvir*).
- **Validation:** Implement FluentValidation or DataAnnotations:
  - NID must be unique.
  - Phone must be valid BD format (start with +880 or 01).

Phase 2: Frontend (React)

- **State Management:** Use Hooks (or Redux/Zustand if preferred) to handle the employee list.
- **UI:** Use a modern library like Tailwind CSS, MUI, or Ant Design.
- **Search Optimization:** Implement a **debounce** (300ms–500ms) on the search bar to optimize API calls.

Phase 3: The SRS Document

Submit a separate file named SRS_Document.md or .pdf. It must include:

- **System Scope:** What the software does (and explicitly what it does *not* do).
- **Entity Relationship Diagram (ERD):** A simple text-based or visual diagram.
- **Edge Cases:** How you handle scenarios like NID duplicates or invalid phone numbers.
- **Assumptions:** Any logic you assumed (e.g., *"An employee can only have one spouse"*).

4. Submission & Evaluation

Submission Steps

1. **GitHub:** Upload your code to a Public Repository.
2. **README:** Your README.md must contain:
   - Steps to run docker-compose (if used) or local Postgres setup.
   - Command to run migrations.
   - Backend and Frontend start commands (e.g., dotnet run and npm run dev).
3. **Email:** Send the repository link and the SRS document to hello@fionetix.com with the subject:
   - *"Fullstack Candidate: [Your Name] - .NET Developer Technical Assessment"*

Evaluation Matrix

Criteria	Weight	Excellent (5/5) Standard
Code Quality	30%	Clean Architecture, proper naming conventions, effective DTO usage.

Functionality	30%	PDF exports work perfectly; Search is fast, accurate, and debounced.
Database Design	20%	Correct relational mapping (Employee ↔ Family).
SRS Clarity	20%	Professional formatting, clear logic, and technical depth.

5. Timeline

- **Deadline:** 7 Mar, 2026

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